

Bioinformatics: Biological Networks

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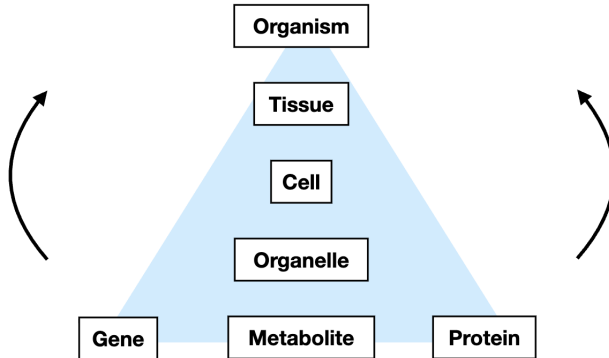
References



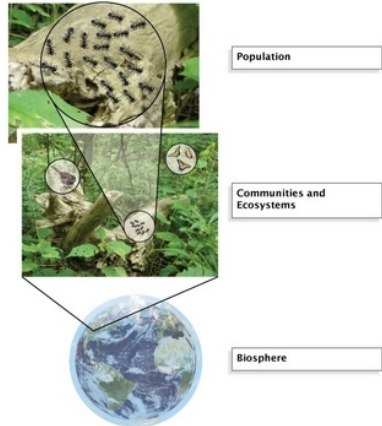
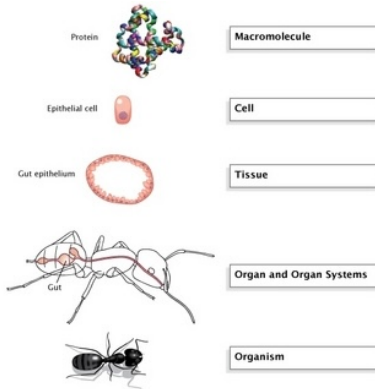
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How to tackle biological complexity?

Moving from reductionist approaches towards global characterisations



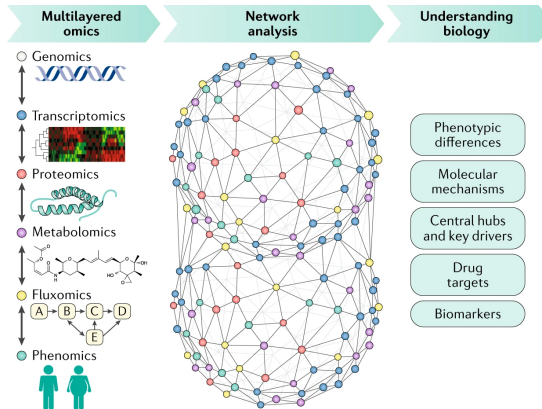
How to tackle biological complexity?



How to tackle biological complexity?

Integrative approaches, and global patterns

- Feature association
- Network analysis
- Modeling



Network Biology

- Biological systems are often represented as networks which are complex sets of interactions or relations between different entities.
- Every biological entity has interactions with other biological entities, from the molecular to the ecosystem level, providing us with the opportunity to model biology using many different types of networks.
- The data explosion that originated in the -omics era of biological research necessitated the development of more systemic approaches to data analysis and a move away from the single gene/protein perspective.
- Network biology allows the representation and analysis of biological systems using tools derived from graph theory.
- Biological network analysis historically originated from the tools and concepts of social network analysis and the application of graph theory to the social sciences.

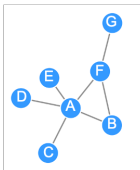


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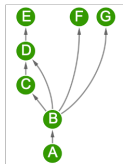
Graph types and edge properties

- The nodes represent different entities (e.g. proteins or genes in biological networks), and edges convey information about the links between the nodes.

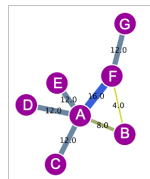
Undirected



Directed



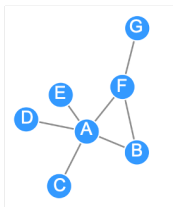
Weighted



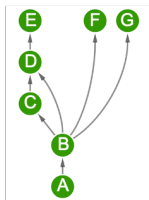
- Undirected edges - Protein Protein Interaction Networks
- Directed edges - metabolic or gene regulation networks. There is a clear flow of signal implied and the network can be organised hierarchically
- Weighted edges - reliability of an interaction, the quantitative expression change that a gene induces over another or even how closely related two genes are in terms of sequence similarity.

Adjacency Matrix

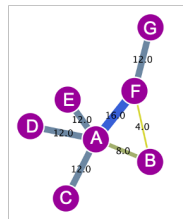
Undirected



Directed



Weighted



	A	B	C	D	E	F	G	Degree
A	0	1	1	1	1	1	0	5
B	1	0	0	0	0	1	0	2
C	1	0	0	0	0	0	0	1
D	1	0	0	0	0	0	0	1
E	1	0	0	0	0	0	0	1
F	1	1	0	0	0	0	1	3
G	0	0	0	0	0	1	0	1

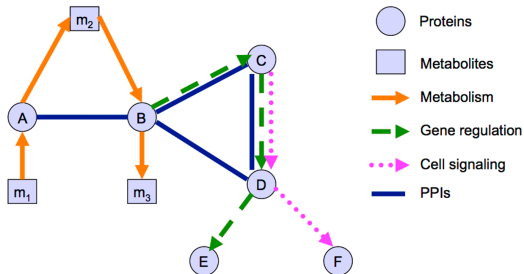
	A	B	C	D	E	F	G	Out-degree
A	0	1	0	0	0	0	0	1
B	0	0	1	1	0	1	1	4
C	0	0	0	1	0	0	0	1
D	0	0	0	0	1	0	0	1
E	0	0	0	0	0	0	0	0
F	0	0	0	0	0	0	0	0
G	0	0	0	0	0	0	0	0

	A	B	C	D	E	F	G	Degree
A	0	8	12	12	12	16	12	72
B	8	0	0	0	0	4	0	12
C	12	0	0	0	0	0	0	12
D	12	0	0	0	0	0	0	12
E	12	0	0	0	0	0	0	12
F	16	4	0	0	0	0	12	44
G	12	0	0	0	0	12	0	24

Adjacency matrices

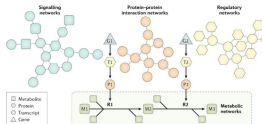
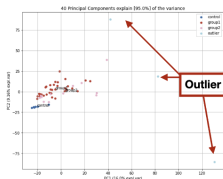
Types of Biological Networks

- Protein-protein interaction networks
- Metabolic networks
- Genetic interaction networks
- Gene / transcriptional regulatory networks
- Cell signalling networks

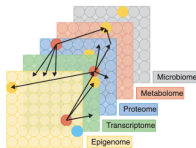
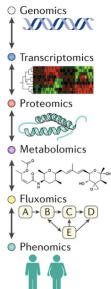


- Sources: Manual curation of scientific literature, High-throughput datasets, Computational predictions, Literature text-mining,

Building Networks



Raw → Pre-processing → Distance calculation → Graph analysis



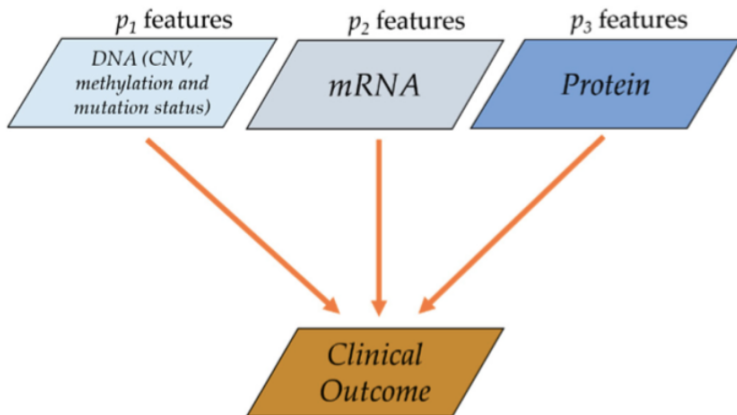
Networks may be build for individual omics or for their integration



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Association Analysis

- Balanced dataset for group sizes
- Common approach:
 - compute correlations between different features (Spearman, Pearson)



Protein Protein Interaction Networks

Protein-protein interaction networks (PPIN) are mathematical representations of the physical contacts between proteins in the cell. These contacts:

- are specific
- occur between defined binding regions in the proteins
- have a particular biological meaning (i.e., they serve a specific function)

Knowledge of PPIs can be used to:

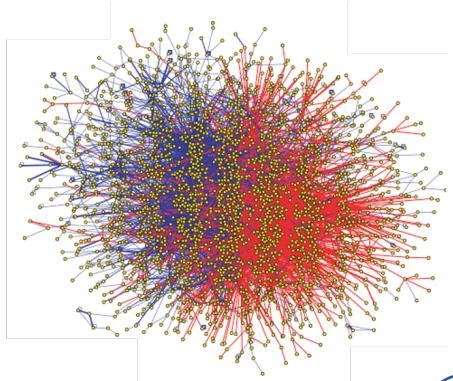
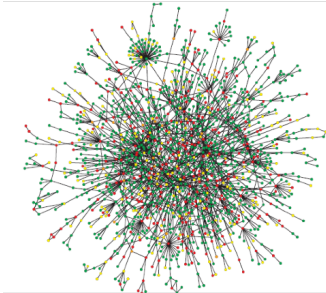
- assign putative roles to uncharacterised proteins
- add fine-grained detail about the steps within a signalling pathway
- characterise the relationships between proteins that form multi-molecular complexes such as the proteasome



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The Interactome

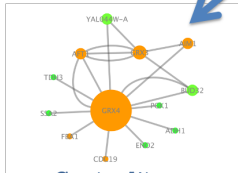
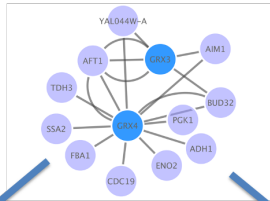
- The interactome is the totality of PPIs that happen in a cell, an organism or a specific biological context.



Yeast (left) and human (right) interactomes obtained using the yeast-two hybrid method. Jeong et al. Nature 2001. 411 (3) and Rual et al. Nature 2005: 437 (4)

Topological Analysis

Base PPI network



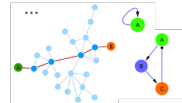
Centrality
analysis



Topological
clustering

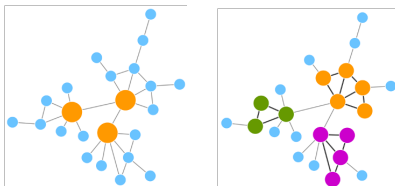
Others:

- Shortest paths
- Motif search (directed networks)



Network Topology

- **The degree of a network** – The degree is the number of edges that connect to a node.
- **Shortest paths** – Shortest paths, or the shortest distance between any two nodes, is used to model how information flows.
- **Scale-free networks** – In scale-free networks most of the nodes are connected to a low number of neighbours and there are a small number of high-degree nodes (hubs) that provide high connectivity to the network.
- **Transitivity** – Transitivity relates to the presence of tightly interconnected nodes in the network called clusters or communities.
- **Centralities** – Centrality can be measured for nodes and for edges and gives an estimation on how important that node/edge is for the connectivity or the information flow of the network.



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Transitivity

- The transitivity or clustering coefficient of a network is a measure of the tendency of the nodes to cluster together.
 - High transitivity means that the network contains communities or groups of nodes that are densely connected internally.
 - In biological networks, finding these communities is very important, because they can reflect functional modules and protein complexes
- A module is an exchangeable functional unit.
 - They are self-contained components of a system with well-defined interfaces with other components.
 - The defining feature of a module is that its intrinsic functional properties do not change when it is placed in a different context.
 - Modules help reduce the complexity of biological networks by giving us a set of reducible, functional units that can be studied as an integrated entity.



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Clustering Coefficient

- How likely is it that two connected nodes are part of a highly connected group of nodes?
- If node v_1 is connected with v_2 and v_3 , it is very likely that v_2 and v_3 are also connected.
- Takes into account degree of a node and the degree of its 1st neighbours

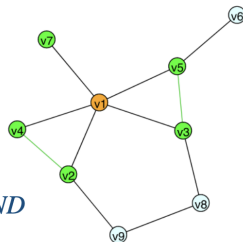
$$C_i = \frac{2 \cdot n}{k_i \cdot (k_i - 1)}$$

- $\deg(v_1) = k = 5$

- n connections between 1st neighbours of $v_1 = 2$

$$C_i = \frac{2 \cdot n}{k_i \cdot (k_i - 1)}$$

$$C(v_1) = \frac{2 \cdot 2}{5 \cdot 4} = 0.2 \quad C(v_7) = \frac{2 \cdot 0}{1 \cdot 0} = 0 \text{ or } ND$$



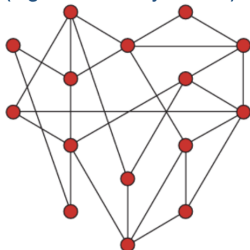
- The global clustering coefficient $C(G)$ is simply the average of its clustering coefficients



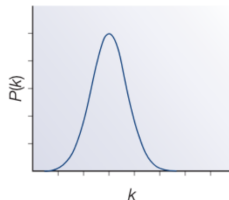
What distinguishes biological networks from random?

- Degree distributions allow us to compare network organization

Random network
(e.g. Erdős-Rényi model)



Poisson degree distribution
shows no highly connected nodes

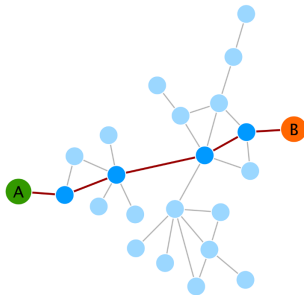


Most nodes have near $\langle k \rangle$

- $C(k)$ shows inverse relationship with k
- Sparsely connected nodes are part of highly modular areas
- Communication between highly clustered neighbourhoods maintained by a few hubs

The small world effect

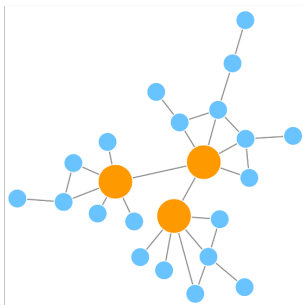
- Protein-protein interaction networks show a small world effect meaning that there is great connectivity between proteins.
- Network's diameter (the maximum number of steps separating any two nodes) is small, no matter how big the network is.
- This level of connectivity has important biological consequences, since it allows for an efficient and quick flow of signals within the network.
- if the network is so tightly connected, why don't perturbations in a single gene or protein have dramatic consequences for the network?



Scale-free networks

- small number of nodes with high degree (the hubs) and a large number of nodes with a low degree

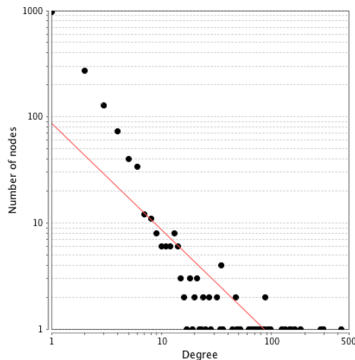
A



Scale-free network

B

Scale-free network degree distribution



Scale-free networks

- Stability

- If failures occur at random, and the vast majority of proteins are those with a small degree of connectivity, the likelihood that a hub would be affected is small
- If a hub-failure occurs, the network will generally not lose its connectedness, due to the remaining hubs

- Invariant to changes of scale

- No matter how many nodes or edges the network has, its properties remain stable
- The presence of hubs is what allows for the small-world effect to be present regardless of the size of the network

- Vulnerable to targeted attack

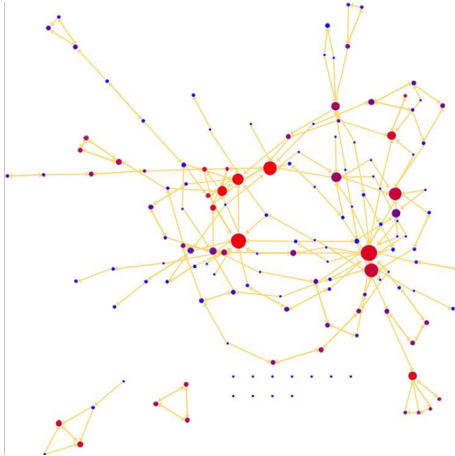
- If we lose a few major hubs from the network, the network is turned into a set of rather isolated graphs
- Hubs are enriched with essential/lethal genes. For example, many cancer-linked proteins are hub proteins (e.g. the tumour suppressor protein p53)



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Centrality Analysis

- Which protein is the most important and why?



- degree, betweenness, closeness, random walk, etc

Closeness centrality

- Closeness centrality measures how short the shortest paths are from node i to all nodes. It is usually expressed as the normalised inverse of the sum of the topological distances in the graph

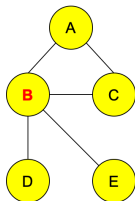
$$CC(i) = \frac{N-1}{\sum_j d(i,j)}$$

where

$i \neq j$,

d_{ij} is the length of the shortest path between nodes i and j in the network,

N is the number of nodes.



	A	B	C	D	E
A	0	1	1	2	2
B	1	0	1	1	1
C	1	1	0	2	2
D	2	1	2	0	2
E	2	1	2	2	0

farness

$$\sum_{j=1}^n d(i,j)$$

$$CC(i) = \frac{N-1}{\sum_j d(i,j)}$$

$$6 \quad (5-1)/6 = 0.67$$

$$4 \quad 1.00$$

$$6 \quad 0.67$$

$$7 \quad 0.57$$

$$7 \quad 0.57$$

$N = 5$ (# of nodes)



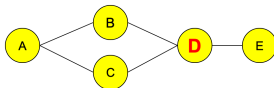
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Betweenness centrality

- Betweenness centrality is based on communication flow.
- Nodes with a high betweenness centrality are interesting because they lie on communication paths and can control information flow. These nodes can represent important proteins in signalling pathways and can form targets for drug discovery.
- defined as the number of shortest paths in the graph that pass through the node divided by the total number of shortest paths.

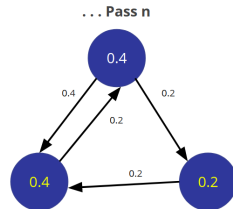
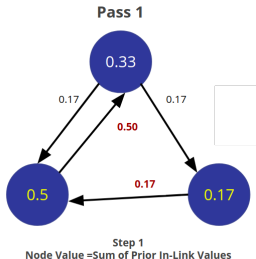
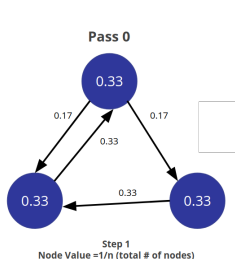
$$C_B(n_i) = \sum_{j < k} g_{jk}(n_i) / g_{jk}$$

Where g_{jk} = the number of geodesics (shortest paths) connecting j,k , and $g_{jk}(n_i)$ = the number that node i is on.



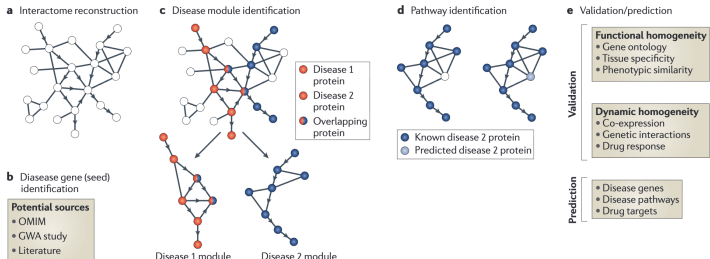
Random Walk - Page Rank

- It works by counting the number and quality of links to a node to estimate its significance.
- The underlying assumption is that a node is only as important as the nodes that link to it.



Modularity

- Modules are physically or functionally associated nodes that work together to achieve a distinct function
- **Hypothesis:** common functional properties (diseases, biological processes, etc.) are associated with the same module
- **Prediction:** in silico, relies on available knowledge
- **Validation:** experimental responses



Modularity

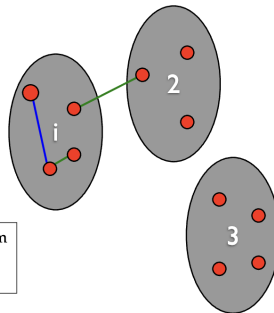
- Modularity is a property of the network that measures the tendency of a graph to be organised into modules
- Modules computed by comparing probability that an edge is in a module vs what would be expected in a random network

e_{ii} = % edges in module i

$e_{ii} = |\{(u,v) : u \in V_i, v \in V_i, (u,v) \in E\}| / |E|$

a_i = % edges with at least 1 end in module i

$a_i = |\{(u,v) : u \in V_i, (u,v) \in E\}| / |E|$



Modularity is:

$$Q = \sum_{i=1}^k (e_{ii} - a_i^2)$$

probability a random edge would fall into module i

probability edge is in module i

High modularity $\Rightarrow$ more edges within the module that you expect by chance.

Modules

- A module (or community) is a set of nodes with a lot of internal connections, but fewer external connections. How to identify modules?
- Maximise Q

Brute-force approach

- 1 Start with 1 node/module
- 2 Compute distances between nodes
- 3 Join closest node
- 4 Re-compute distances between a $2n$ module and each $1n$ module
- 5 Join them if Q increases



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Module detection: Louvain algorithm

Phase 1: greedy modularity optimisation

- 1 Start with 1n/community
- 2 Compute Q by moving i to the community of j
- 3 If $\Delta Q > 1$, node is placed in community
- 4 Repeat 1-3 until no improvement is found. Ties solved arbitrarily

Phase 2: coarse grained community aggregation

- 5 Link nodes in a community into single node.
- 6 Self loops show intra-community associations
- 7 Inter-community weights kept
- 8 Repeat phase 1 on new network



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Louvain

